



MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

WEEK 5 TEAM PROGRESS REPORT

Agent Architecture and Bounded Autonomy

PROJECT: University Student-Support Case Agent

28 September – 2 October 2026

TEAM MEMBERS:

NAME	ROLE
Kabanda Arthur	Project / Requirements Lead
Tumukunde Kato Andrew	Application / Integration Lead
Ariko Sossy Joel	AI Engineering Lead
Garanga John	Quality / Security & DevOps Lead (assembled this report)

1. Deliverables

Deliverable	Status	Evidence
Agent Task Contract	Present; out of date with the merged code	docs/architecture/Agent_Task_Contract.pdf
Agent architecture diagram	Present; out of date with the merged code	docs/architecture/agent_architecture_week_5.png
/chat routed through the bounded loop	Done	PR #19 , src/main.py, src/orchestrator.py
Prompt for the Week 5 workflow	Done as a new version, v4.0	PR #20 , prompts/v4.0.md
Application-level traces	Done; four captured	PR#21
Focused tests	Pass	PR#21 , evidence/traces/week5/pytest-run-garanga.txt

2. Contributions

This records what each member did, including work beyond the Week 5 plan.

Member	Contribution	Evidence
Kabanda Arthur	Wrote the Agent Task Contract and the Week 5 architecture diagram; merged PRs #19 and #20.	d451918
Tumukunde Kato Andrew	Extended the orchestrator into the bounded Sense/Plan/Act/Observe/Stop loop and routed /chat through it.	37e87a8 , e6f3fa8 , PR #19
Ariko Sossy Joel	Wrote prompt v4.0 and made it the default. Beyond the plan: hardened the stop-condition logic in the orchestrator, added tests/test_stop_conditions.py, captured scripted traces S01–S05, and wrote an individual report.	8f76cc8 , b88f749 , PR #20
Garanga John	Verified boundary and failure behaviour, captured four live traces through /chat, ran the sanitization check, and assembled this report.	PR#21

3. Execution traces

Each trace is a single POST /chat request to the running app with a live Gemini model and an invented student number. Details are in the trace index, evidence/traces/week5/chat/README.md.

Trace	Request	Tools run	How it ended
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Trace	Request	Tools run	How it ended
C01 success	Raise a ticket for an exam clash	retrieve_evidence, create_support_ticket	goal_satisfied; ticket OPEN. The approval gate did not fire (Section 5, issue 3).
C02 failure / recovery	Check an unknown case; if missing, raise a ticket	get_case_status (CASE_NOT_FOUND), retrieve_evidence, create_support_ticket	goal_satisfied; ticket OPEN.
C03 round limit	Same as C02 with MAX_TOOL_ROUND=1	get_case_status (CASE_NOT_FOUND)	round_limit_reached; reply says the limit was reached, no ticket created or claimed.
C04 approval gate	Raise a HIGH urgency examination ticket for the same clash	retrieve_evidence, create_support_ticket	approval_pending; ticket PENDING_STAFF_APPROVAL, reply says it awaits staff review.

C01 and C02 ran on the default gemini-3.6-flash. C03 and C04 ran on gemini-3.5-flash because the default model's free quota of 20 requests a day ran out after C02. The scripted traces S01–S05 remain as loop-level evidence of each stop condition; they do not pass through /chat.

4. Verification

Run	Result	Log
pytest tests/test_orchestrator.py tests/test_stop_conditions.py tests/test_tools.py -v	273 passed, 5 skipped	evidence/traces/week5/pytest-run-garanga.txt
pytest -q (full suite)	313 passed, 5 skipped	evidence/traces/week5/pytest-full-run-garanga.txt

The five skipped tests are live-model tests that are off by default.

Boundaries confirmed by passing tests

- Undeclared tools are refused and not executed.
- The round cap stops the loop and always returns a message.
- Examination and high-urgency tickets stop at the approval gate.
- A tool failure is re-planned once, then the run stops with no invented result.
- Disallowed topics cannot be raised as tickets.
- Missing or malformed fields are rejected.

Sanitization

.env is not tracked and no API keys were found in tracked files. The four new traces contain only the invented student number.

5. Open issues

1. **Student identifiers in committed files.** Four 10-digit student numbers are in the sandbox seed data, and 67 lines across tracked files match the student-number pattern, including the tests and the Week 4 and Week 5 evidence. One seeded number matches a number a team member gives as their own in their individual report, which also states a registration number. A registration-style number is also in tests/test_tools.py and the test logs. The team needs to confirm which are real and replace them with invented ones.
2. **Contract and diagram do not match the code.** They describe two tools, a status check after ticket creation, and v3.0 as the default prompt. The code has three tools (retrieve_evidence was added), stops at the approval gate straight after creating the ticket, and defaults to v4.0. The diagram also says /chat calls the model directly and that repeated identical tool calls are not blocked; both are no longer true.
3. **An exam clash can skip the approval gate (from C01).** Asked in natural wording, the model filed the exam clash as TIMETABLE/MEDIUM, so the ticket went to the general queue as OPEN with no staff approval. The gate fired only when the message stated the category and urgency (C04). The gate depends on how the model labels the request.
4. **No test for the repeated-call rule.** The orchestrator refuses a repeated identical tool call (DUPLICATE_TOOL_CALL), but no test references that code.